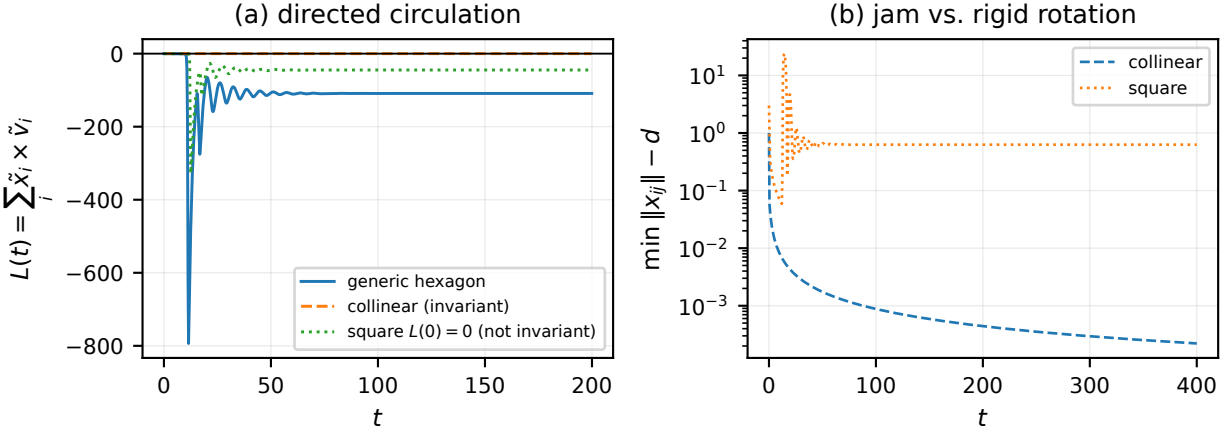
(a) directed circulation
$L(t) = \sum_i \tilde{x}_i \times \tilde{v}_i$
0
−200
−400
−600
−800
0
50
100
150
200
t
generic hexagon
collinear (invariant)
square $L(0) = 0$ (not invariant)
(b) jam vs. rigid rotation
$\min \|x_{ij}\| - d$
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
0
100
200
300
400
t
collinear
square